

Task 4: Design for Testability (DFT) & Basic Physical Verification Report

1. Project Overview

- **Design Name:** 8-bit Registered Arithmetic Logic Unit (alu_8bit)
- **Target Software:** AMD Vivado Design Suite 2026.1
- **Target Device:** Kintex-7 FPGA (xc7vx485tffg1157-1)
- **Objective:** Implement Design for Testability (DFT) using internal signal marking and perform Design Rule Checking (DRC) physical verification.

2. HDL Source Code with DFT Markings (alu_8bit.v)

Verilog

```
module alu_8bit (  
    input wire clk,  
    input wire [7:0] A_in,  
    input wire [7:0] B_in,  
    input wire [2:0] opcode_in,  
    output reg [7:0] result,  
    output reg zero_flag  
);  
  
    // Input registers with DFT Markings for internal observability  
    (* mark_debug = "true" *) reg [7:0] A;  
    (* mark_debug = "true" *) reg [7:0] B;  
    (* mark_debug = "true" *) reg [2:0] opcode;  
  
    always @(posedge clk) begin  
        A <= A_in;  
        B <= B_in;  
        opcode <= opcode_in;  
    end  
  
    // Combinational ALU logic core  
    (* mark_debug = "true" *) reg [7:0] alu_out;  
    always @(*) begin
```

```

    case (opcode)
        3'b000: alu_out = A + B;
        3'b001: alu_out = A - B;
        3'b010: alu_out = A & B;
        3'b011: alu_out = A | B;
        3'b100: alu_out = A ^ B;
        default: alu_out = 8'b0;
    endcase
end

// Output registers
always @(posedge clk) begin
    result <= alu_out;
    zero_flag <= (alu_out == 8'b0);
end
endmodule

```

3. Physical Constraints File (constraints.xdc)

```

Tcl

# Primary Clock Constraint
create_clock -period 12.000 -name sys_clk [get_ports clk]

# Input Delay Constraints relative to primary clock
set_input_delay -clock sys_clk 2.000 [get_ports {A_in[*] B_in[*] opcode_in[*]}]

# Output Delay Constraints relative to primary clock
set_output_delay -clock sys_clk 2.000 [get_ports {result[*] zero_flag}]

```

4. DFT & Physical Verification Summary Table

Verification Parameter	Identified Status / Value	Assessment
Marked Debug Signals	A (8), B (8), opcode (3), alu_out (8)	PASSED
Total Probed Debug Nets	27 Unassigned Debug Nets	PASSED
DRC Critical Errors	0 Critical Errors	PASSED
DRC Critical Warnings	2 Critical Warnings (NSTD-1, UCIO-1)	REVIEWED
I/O Pin Assignment Status	Unconstrained package pins (Default IOSTANDARD)	EXPLAINED

5. DFT & Physical Verification Analysis

- Design for Testability (DFT) Strategy:** Internal registers (A, B, opcode) and intermediate combinational buses (alu_out) were marked using the (* mark_debug = "true" *) synthesis attribute. Synthesis successfully preserved all 27 debug nets without gate optimization, making internal nodes accessible for logic analyzer probing. In ASIC test flows, this corresponds to replacing standard flip-flops with scan flip-flops to build scan chains for detecting physical stuck-at-0 and stuck-at-1 faults.
- Design Rule Check (DRC) Execution:** Running Report DRC on the synthesized netlist generated zero critical errors and flagged expected I/O pin planning warnings.
- DRC Violation Analysis:** The NSTD-1 critical warning indicates that 29 logical ports are using default I/O standards because exact physical FPGA pin locations were not locked down in the .xdc file. This is standard during baseline synthesis verification and is resolved prior to bitstream generation by binding ports to specific package pins (e.g., LVCMOS33).

6. Screenshots & Deliverables

Image 1: Synthesized Gate-Level Schematic

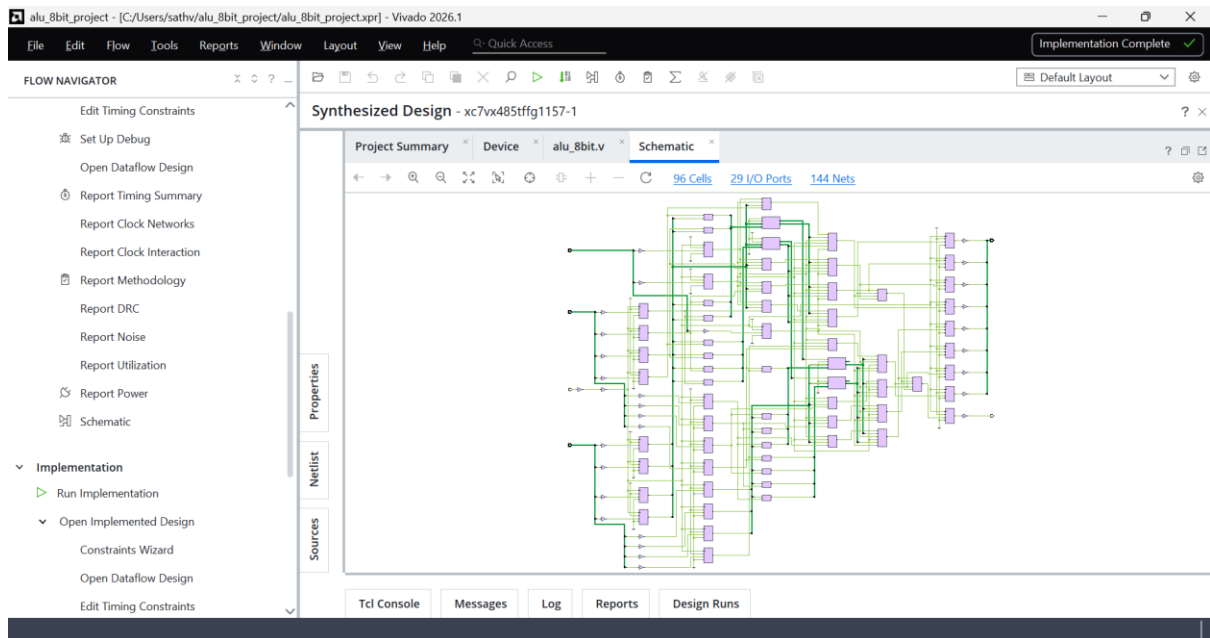


Image 2: DFT Insertion & Probed Debug Nets

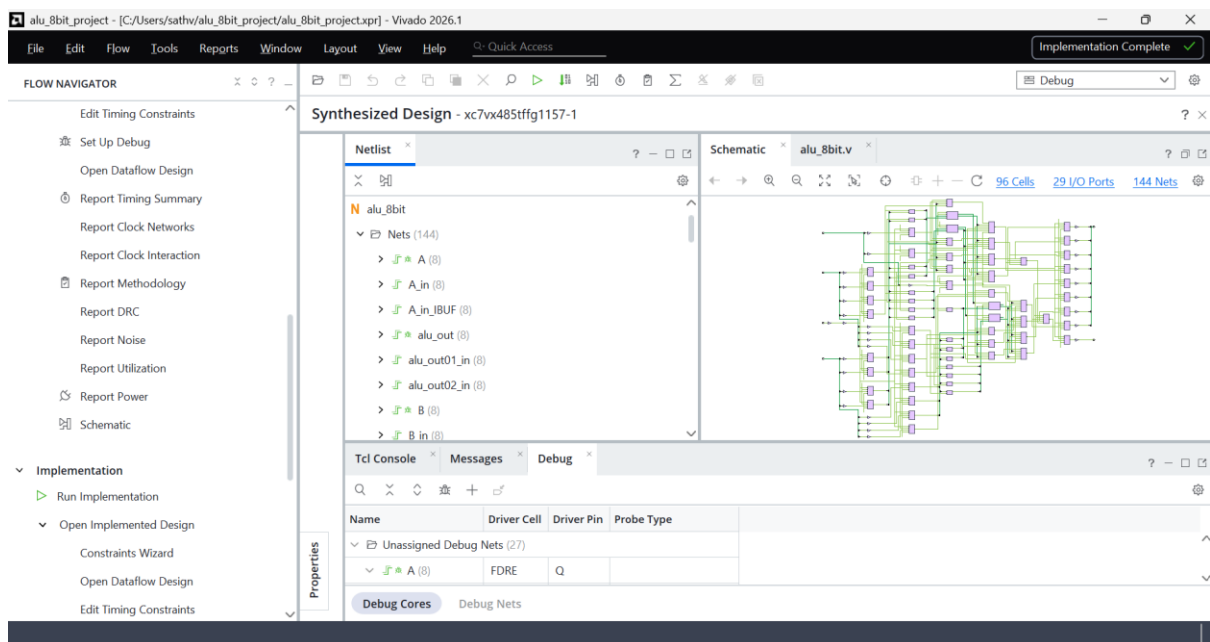


Image 3: Report DRC Verification Results

FileEditFlowToolsReportsWindowLayoutViewHelpQuick AccessImplementation Complete

FLOW NAVIGATOR

Constraints Wizard

Edit Timing Constraints

Set Up Debug

Open Dataflow Design

Report Timing Summary

Report Clock Networks

Report Clock Interaction

Report Methodology

Report DRC

Report Noise

Report Utilization

Report Power

Schematic

Implementation

Run Implementation

Open Implemented Design

Constraints Wizard

Open Dataflow Design

Synthesized Design - xc7vx485tffg1157-1

Netlist

Schematic

al_u_8bit.v

Tcl Console

Messages

DRC

Debug

2 Critical Warnings

1 Warning

Hide All

Name	Severity	Details
All Violations (3)		
Pin Planning (3)		
NSTD-1 (1)		
NSTD #1	Critical Warning	29 out of 29 logical ports use I/O standard (IOSTANDARD) value "DEFAULT", instead of a user assigned specific value. This may cause I/O contention or incompatibility with the board power or connectivity affecting performance, signal integrity or in extreme cases cause damage to the device or the components to which it is connected. To correct this violation, specify all I/O standards. This design will fail to generate a bitstream unless all logical ports have a user specified I/O standard value defined. To allow bitstream creation with unspecified I/O standard values (not recommended), use this command: set_property SEVERITY {Warning} [get_drc_checks NSTD-1]. NOTE: When using the Vivado Runs infrastructure (e.g. launch_runs Tcl command), add this command to a .tcl file and add that file as a pre-hook for write_bitstream step for the implementation run. Problem ports: A_in[7:0], B_in[7:0], opcode_in[2:0], result[7:0], clk, zero_flag.

drc_1 (3 violations)